

Description

Circuit Builder specific API calls.

Remarks

AutoCAD Electrical includes these calls that are specifically set up to work with the "Circuit Builder" feature. Encode these calls in the spreadsheet that drives this tool.

Topics

Name	Description
c:ace cb adjust_wire_crossings	Process wire found at location "xy" and adjust any wire crossings to match active drawing's wire cross graphics setting (nth 49 GBL_wd_m).
c:ace cb anno	This data passed as argument "data" which references a global (held in memory) list of data lists. This does not process the first two sublists in this global data structure but starts at the third list (index=2, catalog assignment information begins. Data structure is illustrated below. If target catalog assignmer RATINGx attributes are present and non-blank values are present in the "data" global variable, then it will annotate on to the component's attributes.
c:ace cb anno2	
c:ace cb attr	Push attribute value "atval" out to attribute name "atnam" on block insert that is carried as first eleme #hdl_list
c:ace cb catalog_lookup	Open the catalog lookup dialog box.
c:ace cb eval	This function can be encoded into the Circuit Builder spreadsheet and greatly expands its ability to enc lisp expressions into the spreadsheet.
c:ace cb figure_mccdb_fname	Internal Circuit Builder use. Look for ace_electrical_standards.mdb file to use for the current project. 1st choice= ace_electrical_standards.mdb in same subdir as project's wdp file 2nd choice= ace_electrical_standards.mdb in USER subdir 3rd choice= ace_electrical_standards.mdb in catalogs subdir 4th choice= ace_electrical_standards.mdb in ACAD paths
c:ace cb get	Retrieve "codename" attribute value or component handle number previously saved by a call to (c:ace_cb_save...) in global GBL_wd_cb_save_@list@ during the current or last circuit build. This glot at the start of the Circuit Build command. This means that it is still in memory after a circuit is comple before a new one is started) and may be accessed by some post-processing application.
c:ace cb insert	Invoke Circuit Builder and insert complete circuit without any user prompts.
c:ace cb insym	Insert component, break underlying wires.
c:ace cb insym_DynamicBlock	Inserts dynamic block component and adjusts to hit the marked wire connections.
c:ace cb insym_signal	Inserts a "signal" arrow symbol at the marker block location.
c:ace cb multipole	Insert mult-pole component, break underlying wires.
c:ace cb nested	Circuit Builder use. Insert nested circuit "fname" at location "xy". Circuit is inserted "exploded". Scale is from (nth 1 GBL_wd_cb_data) global or defaults to 1.0 if global not found.
c:ace cb parametric_connector	Insert parametrically generated connector component.
c:ace cb rung_spacing	Retrieve horizontal and vertical bus spacing defaults from (nth 4 GBL_wd_cb_data) and (nth 3 GBL_w respectively. Finds circuit builder marker block CODE/ORDER value pair being processed from this glot (car GBL_wd_cb_marker_data) (cadr GBL_wd_cb_marker_data)) and then all other marker blocks in l by match on this CODE/ORDER combination.
c:ace cb save	Extract attribute value, pair it with an alias codename value and handle value, and hold in memory for retrieval later by call to (c:ace_cb_get...). Codename/Value pairs are held in memory in an AutoLISP ("GBL_wd_cb_save_@list@". The structure of this global is (list (list codename attrval hdl)(list codenar hdl)...))
c:ace cb save_for_link	Use in the spreadsheet right after one of the insert Circuit Builder APIs. Blocks grouped together for li linked with the dashed link line after the entire circuit is built."
c:ace cb save_selection	Push circuit selection data out to the main motor or load component of the circuit being processed. Th allows the circuit to be edited to re-display / re-calculate wire sizes or referenced during another circu (example: reference an existing one-line circuit when inserting an equivalent three-phase schematic c
c:ace cb set_wiretype	Extract wire size from global (i.e. held in memory) list of data and format an appropriate wire type lay
c:ace cb stretch_wire_connect	Find wire under "xy" point and try to stretch to connect to nearest crossing wire or component wire co point.
c:ace cb trim	Trim wire found at location "xy".
c:ace cb wnum	Generate and insert a wire number on target wire network line segment "wen" or on line segment und coordinate "passed_xy".

AutoCAD Electrical 2025: AutoLISP Reference

c:ace_cb_adjust_wire_crossings

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Description

Process wire found at location "xy" and adjust any wire crossings to match active drawing's wire crossing graphics setting (nth 49 GBL_wd_m).

AutoLISP

(c:ace_cb_adjust_wire_crossings xy options)

Parameters

Parameters	Description
xy	nil = use coordinate of current marker block being processed coordinate = process underlying wire at this coordinate non-blank string = process all marker block(s) that are marked with this string as attribute value (wild card match) AND have same ORDER value as the main marke being processed
options	future

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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_cb_anno

[Example](#) [Section L - Circuit Builder](#) [Send Feedback](#)

Description

This data passed as argument "data" which references a global (held in memory) list of data lists. This function does not process the first two sublists in this global data structure bu the third list (index=2) where catalog assignment information begins. Data structure is illustrated below. If target catalog assignment and/or RATINGx attributes are present and noi values are present in the "data" global variable, then the values will annotate on to the component's attributes.

AutoLISP

(c:ace_cb_anno data options)

Parameters

Parameters	Description
data	global (held in memory) data structure passed from circuit builder's 2nd ARX GUI. argument set to nil then it will default to using global "data" if exists.
options	1's bit set= don't hide repeated RATING1 on child poles clear= hide repeated RATING1 values on child poles 2's bit set (and 1's bit clear)= show last RATING1, hide the rest clear (and 1's bit c first RATING1, hide the rest 4's bit set=multi-catalog RATINGx text values inserted "as defined" clear= multi-catalog RATING flag increment to next higher unused RATINGx attrib. For example, both main and 1st multi-cat entries have non-blank RATING1=xx val main goes to RATING1, the multi-cat goes to RATING2 (ex: for a fused-disconnect RATING1 for switch amps and the 2nd catalog number for a fuse gets pushed up to 8's bit set= main part number copied to child poles if they carry MFG/CAT attribute clear= main part number does NOT copy to child poles 16's bit set= any extra part number copied to child poles clear= extra part number does NOT copy to child poles

Example

How this works:

```

; The "data" structure is this (note that the first two sublists are not processed by this
function):
(list
;      (list type hp voltage rpm hertz FullLoadAmps) ; or nil if not yet selected
;      (list (wire1 lst) (wire2 lst) (wire2 lst)) ; where wirex lst=(list size laynam flag) or nil
```

```
; ("CB" "AB" "1492-CB1DF005" "" "1-POLE CIRCUIT BREAKER" "0.5A" "" "" "" "" "" "1"))
```

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_anno2
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Description

AutoLISP

```
(c:ace_cb_anno2 data2 code target options param_list)
```

Parameters

Parameters	Description
data2	global (held in memory) data structure passed from circuit builder's 2nd ARX GUI. argument set to nil then it will default to using global "data2" if exists.
code	The "Ax" identifier, used to find a specific sublist in the "data2" structure.
target	The target attribute name to push value out to. If blank, default to RATING1.
options	1's bit set= don't hide repeated RATING1 on child poles clear= hide repeated RATING1 values on child poles 2's bit set (and 1's bit clear)= show last RATING1, hide the rest clear (and 1's bit c first RATING1, hide the rest 4's bit set=multi-catalog RATINGx text values inserted "as defined" clear= multi-catalog RATING flag increment to next higher unused RATINGx attrib. For example, both main and 1st multi-cat entries have non-blank RATING1=xx val main goes to RATING1, the multi-cat goes to RATING2 (ex: for a fused-disconnect RATING1 for switch amps and the 2nd catalog number for a fuse gets pushed up to 8's bit set= main part number copied to child poles if they carry MFG/CAT attribute clear= main part number does NOT copy to child poles 16's bit set= any extra part number copied to child poles clear= extra part number does NOT copy to child poles
param_lst	nil (future)

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_attr
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Description

Push attribute value "atval" out to attribute name "atnam" on block insert that is carried as first element in global #hdl_list

AutoLISP

```
(c:ace_cb_attr atnam atval formatstr options)
```

Parameters

Parameters	Description
------------	-------------

options

1's bit set = prepend "atval" to any existing value found on the attribute (atval is t
2's bit set = append "atval" to any existing value found on the attribute (atval beco
suffix)
4's bit set = overwrite existing value even if "atval" is blank bit clear means preser
existing value if "atval" is blank (i.e. do not overwrite existing value with a blank v
8's bit set = write to attribute or out to Xdata if attribute not present

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_catalog_lookup
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Description

Open the catalog lookup dialog box.

AutoLISP

(c:ace_cb_catalog_lookup table_code mfg cat asmb options)

Parameters

Parameters	Description
table_code	table name like "SS" or "PB" or "TRMS" or "PLCIO" mfg cat asmb
mfg	manufacturer code (or nil)
cat	catalog part number (or nil)
asmb	ASSYCODE value (or nil)
options	1's bit set = return is list of lists including any subassy entries ; nil or 1's bit not se is one list of the main part number

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_eval
[Example 1](#) [Example 2](#) [Section L - Circuit Builder](#) [Send Feedback](#)

Description

This function can be encoded into the Circuit Builder spreadsheet and greatly expands its ability to encode full lisp expressions into the spreadsheet.

AutoLISP

(c:ace_cb_eval str)

Parameters

Parameters	Description
str	Lisp expression

Example 1

```
(c:ace_cb_eval "(setq overloads 1)")
```

Example 2

```
(c:ace_cb_eval "(if (= overloads 1)(c:ace_cb_insym #xyz nil 'HOL22' #scl 8 nil)))")
```

In this example, the block name HOL22 is shown with single quotes instead of double-quote characters.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_figure_mccdb_fnam
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Description

Internal Circuit Builder use. Look for ace_electrical_standards.mdb file to use for the current project.

- 1st choice= ace_electrical_standards.mdb in same subdir as project's wdp file
- 2nd choice= ace_electrical_standards.mdb in USER subdir
- 3rd choice= ace_electrical_standards.mdb in catalogs subdir
- 4th choice= ace_electrical_standards.mdb in ACAD paths

AutoLISP

```
(c:ace_cb_figure_mccdb_fnam)
```

Returns

Full path name to target file. nil=failure.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_get
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Description

Retrieve "codename" attribute value or component handle number previously saved by a call to (c:ace_cb_save...) in global GBL_wd_cb_save_@list@ during the current or last circu global is reset at the start of the Circuit Build command. This means that it is still in memory after a circuit is completed (and before a new one is started) and may be accessed by s processing application.

AutoLISP

```
(c:ace_cb_get codename options)
```

Parameters

Parameter	Description
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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_cb_insert

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Description

Invoke Circuit Builder and insert complete circuit without any user prompts.

AutoLISP

```
(c:ace_cb_insert param_lst)
```

Parameters

Parameters	Description
param_lst	(list xy template_dwg xls_fnam xls_sheet insckt_scl symscsl rungvertdist runghoriz history_name options annoSpecifics) where "xy" = main template insertion coordinate, nil=user prompted to pick insertion poi "template_dwg" = main template dwg to insert; ".DWG" extension is not required; AutoCAD Electrical and ACAD paths are searched if the full path is not provided. "xls_fnam" = circuit builder XLS file, nil = default to "ace_circuit_builder.xls" Norm Electrical and ACAD paths are searched if the full path is not provided. "xls_sheet" = code sheet that corresponds to the selected template "insckt_scl" = main template insertion scale, nil = use drawing default "symscsl" = component insertion scale, nil = use current default "rungvertdist" = spacing adjustment distance between marked horiz wires "runghorizdist" = spacing adjustment distance between marked vertical wires "history_name" = nil if no "history" reference, otherwise = previously assigned circ "options" = 1's bit set=debug messages on, nil=no options selected "annoSpecifics" = optional annotation specifics, in the format of list of list of "mark attribute name" and "attribute value" to be applied to the inserted circuit.

Example 1

Insert 3-phase non-reversing motor starter circuit as defined in the default ace_circuit_builder.xls spreadsheet file. Insert at coordinate 5.0,23.5 on the active drawing.

```
(c:ace_cb_insert (list "5.0,23.5,0.0" "ace_cb1_FVNR_H" nil "3PH_H" nil nil nil nil nil))
```

Example 2

To insert similar circuit with specific annotations applied.

```
(c:ace_cb_insert (list "5.0,23.5,0.0" "ace_cb1_FVNR_H" nil "3PH_H" nil nil nil nil nil (list (list "PB01" "DESC1" "Start") (list "PB02" "DESC1" "S
```

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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_cb_insym

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Description

Insert component, break underlying wires.

AutoLISP

xy	nil = use coordinate of current marker block being processed; coordinate = trim ur wire at this; non- blank string = trim user under marker block(s) that are marked ' string as their CODE attribute value (wild card match) AND have same ORDER val main marker block being processed
future	nil (future use)
blknam	block name to insert. Can include full path. Does not require ".dwg" extension.
scl	insertion scale; nil = follows (nth 5 GBL_wd_cb_data), and if nil then defaults to v global GBL_wd_scl or 1.0
options	1's bit set = open INS/EDIT dialog box after insertion 2's bit set = return "handle" of inserted symbol (otherwise return ent name) 4's bit set = when 1's bit = 0, suppress parent component or terminal tag auto-gen 8's bit set = allow motor angled connections to go to non-wire LINE entity (used by BLDR) "H" (or 2 or 8) = horizontal, "V" (or 1 or 4) =vertical
anno_string	optional annotation to assign attribute values and/or lisp expressions to execute. S each value with a semi-colon. For example: DESC1=MOTOR;RATING5=30A

Returns

nil if fail, otherwise return entity name or handle of inserted component

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_insym_DynamicBlock
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Description

Inserts dynamic block component and adjusts to hit the marked wire connections.

AutoLISP

```
(c:ace_cb_insym_DynamicBlock xy future blknam scl options anno_string)
```

Parameters

Parameters	Description
xy	nil = use coordinate of current marker block being processed coordinate = trim underlying wire at this non-blank string = trim user under marker block(s) that are marked with this strin CODE attribute value (wild card match) AND have same ORDER value as the main block being processed.
future	nil (future use)
blknam	The block name to insert. Can include full path. Does not require ".dwg" extension.
scl	insertion scale, nil = follows (nth 5 GBL_wd_cb_data), and if nil then defaults to v global GBL_wd_scl or 1.0
options	1's bit set = display INS/EDIT dialog after insertion 2's bit set = return "handle" of inserted symbol (otherwise return ent name) 4's bit set = when 1's bit =0, suppress parent component or terminal tag auto-gen 8's bit set = allow motor angled connections to go to non-wire LINE entity (used by BLDR)
anno_string	optional annotation to assign attribute values and/or lisp expressions to execute. S each value with a semi-colon. For example: DESC1=MOTOR;RATING5=30A

Returns

nil if fail, otherwise returns entity name or handle of inserted component.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_insym_signal
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Description

Inserts a "signal" arrow symbol at the marker block location.

AutoLISP

```
(c:ace_cb_insym_signal xy future blknam scl options anno_string)
```

Parameters

Parameters	Description
xy	nil = use coordinate of current marker block being processed coordinate = trim underlying wire at this non-blank string = trim user under marker block(s) that are marked with this strin CODE attribute value (wild card match) AND have same ORDER value as the main block being processed.
future	nil (future use)
blknam	block name to insert (nil=auto-calculate block name). Can include full path. Does r ".dwg" extension.example: "HA1S1"
scl	insertion scale, nil = follows (nth 5 GBL_wd_cb_data), and if nil then defaults to vz global GBL_wd_scl or 1.0
options	1's bit set = display INS/EDIT dialog after insertion 2's bit set = return "handle" of inserted symbol (otherwise return ent name) 4's bit set = when 1's bit =0, suppress parent component or terminal tag auto-gen
anno_string	optional annotation attribute= (ex: "@SIGCODE@") - value to insert and/or lisp ex to execute

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_multipole
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Description

Insert mulit-pole component, break underlying wires.

AutoLISP

```
(c:ace_cb_multipole xy default_dir blknam_lst polecnt scl options anno_string)
```

Parameters

Parameters	Description
default_dir	build direction 8 = build down, 2 = build up, 1 = build left to right, 4 = build right nil = build down if horiz wire, right if vertical wire. If the specific marker block that this call carries a ".D=[num]" entry on its MISC1 attribute, then this MISC1 value any passed "default_dir" value. 4's bit set = allow connections to LINEs that are not (yet) on a valid wire layer
blknam_lst	parent symbol name, or (list parent_blknam child_blknam) = list of parent name a name or child_blknam if inserting multipole child component
polecnt	total number of poles to insert

nil, then defaults to drawing default. If the specific marker block that triggers this is a MISC1 value with a tag format value, then the marker block's value overrides both argument value and the drawing default tag format value.

anno_string	optional annotation to assign attribute values and/or lisp expressions to execute. See each value with a semi-colon. For example: DESC1=MOTOR;RATING5=30A
-------------	--

Returns

nil if fail, otherwise returns list of entity names or handles of inserted poles, in order. This list also held in AutoLISP global #hdl_list.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_nested
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Description

Circuit Builder use. Insert nested circuit "fnam" at location "xy". Circuit is inserted "exploded". Scale is pulled from (nth 1 GBL_wd_cb_data) global or defaults to 1.0 if global not found.

AutoLISP

```
(c:ace_cb_nested fnam xyz options_lst)
```

Parameters

Parameters	Description
fnam	File name of nested circuit drawing. ".DWG" extension is not required. Normal AutoCAD Electrical and ACAD paths are searched if full path not provided.
xyz	= nil = use coordinate of current marker block being processed = coordinate = insert connector at this XY coordinate = non-blank string = insert at marker block(s) that are marked with this string
options_lst	future

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_parametric_connector
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Description

Insert parametrically generated connector component.

AutoLISP

```
(c:ace_cb_parametric_connector xy direction flip pincont pintext spacing wn_chg scl other_options_lst anno_string)
```

Parameters

Parameters	Description

spacing	1 = fixed pin spacing, 2 = adjust to underlying wire spacing
wn_chg	1 = wire number changes across connector, 0 = no change through connector
scl	nil = default scale (follows global GBL_wd_scl_plcbld)
other_options_lst	future
anno_string	nil for none, otherwise list of attrib = value pairs and/or other override flags

Returns

nil if fail; otherwise list of entity names of connector or connector pieces that inserted

Remarks

This function depends upon a temporary block of data created as Circuit Builder runs. This function will not work if called outside of Circuit Builder.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_rung_spacing
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Description

Retrieve horizontal and vertical bus spacing defaults from (nth 4 GBL_wd_cb_data) and (nth 3 GBL_wd_cb_data) respectively. Finds circuit builder marker block CODE/ORDER value processed from this global: (nth (car GBL_wd_cb_marker_data) (cadr GBL_wd_cb_marker_data)) and then all other marker blocks in the group by match on this CODE/ORDER com

AutoLISP

```
(c:ace_cb_rung_spacing anno_str)
```

Parameters

Parameters	Description
anno_str	future

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_save
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Description

Extract attribute value, pair it with an alias codename value and handle value, and hold in memory for possible retrieval later by call to (c:ace_cb_get...). Codename/Value pairs are memory in an AutoLISP global "GBL_wd_cb_save_@list@". The structure of this global is (list (list codename attrval hdl)(list codename attrval hdl)...)...

AutoLISP

```
(c:ace_cb_save codename attrname ent_lst options)
```

Parameters

	"TAG1*,TAG2,TAGSTRIP,P_TAG1"
ent_lst	List of entities to process (i.e. look for attribute "attrname" and get value) nil then entity held in GBL_wd_cb_markerblock_start_entlast and work until hit end of data
options	1's bit set = extract numeric substring (ex: return "250" if attrval = "M-250") If no numeric part found, then just return the whole attribute value string.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_save_for_link
[Section L - Circuit Builder](#) [Send Feedback](#)

Description

Use in the spreadsheet right after one of the insert Circuit Builder APIs. Blocks grouped together for linking are linked with the dashed link line after the entire circuit is built."

AutoLISP

(c:ace_cb_save_for_link grp place)

Parameters

Parameters	Description
grp	to allow you to define any number of groups of blocks to link within the circuit. Int
place	the order within the group to link together. Integer.

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_save_selection
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Description

Push circuit selection data out to the main motor or load component of the circuit being processed. This data allows the circuit to be edited to re-display / re-calculate wire sizes or r during another circuit insert (example: reference an existing one-line circuit when inserting an equivalent three-phase schematic circuit).

AutoLISP

(c:ace_cb_save_selection hdl #data callback2 options)

Parameters

Parameters	Description
hdl	Handle target component, nil=use global #hdl_list to determine current componen processed. #data : nil = use default #data global, if it exists. This motor/load and data is a list of two lists. An example of the first part of this #data list is shown her ("Induction" "25" "HP" "208" "3" "60" "1200" "76" "1.25" "0" "AWG" "CU" "THWN" "" "" "0" "0" "0.8" "" "0" "" "0" "0.8" "" "" "0.08" "6AWG" "!MCC_CTRL")

value on the "hdl" motor or load symbol.

If no data, then the Xdata tags are not created. #data : nil = use default #data gk exists.
This includes the data that is formatted and pushed on to VIA_WD_CB_CIRCPARAN value. If no data, then the Xdata tag is not created.

callback2	nil = use the default found in (car GBL_wd_cb_select_circuit_callback) and (cadr GBL_wd_cb_select_circuit_callback). This includes the data that is formatted and pushed onto the VIA_WD_CB_CIRCSEI value.
options	nil

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AutoCAD Electrical 2025: AutoLISP Reference
c:ace_cb_set_wiretype
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Description

Extract wire size from global (i.e. held in memory) list of data and format an appropriate wire type layer name.

AutoLISP

(c:ace_cb_set_wiretype data motor_wire_index options misc1_spreadsheet)

Parameters

Parameters	Description
data	global (held in memory) data structure passed from circuit builder's 2nd ARX GUI. argument set to nil then it will default to using global "data" if exists. This "data" is multiple lists with this function accessing only the second list of the overall list, tha 1 #data) or (cadr #data)". This second entry consists of multiple sublists, each sut a conductor feeding the circult's main load (ex: motor). These sublists are in this fi size_substr default_laynam flag) where "size_substr" = wire size pulled from ace_electrical_standards.mdb "default_laynam" = the default wire layer name for the marked conductor "flag" = internal use The size_substr value can be accessed through alias codename "@WSIZE@" by eit MISC1 attribute value on the marker block or as the value for argument "misc1_sp encoded in the spreadsheet entry. The format is preceeded by a "_WIRETYPE=" pr example, "_WIRETYPE=BRN-@WSIZE@-AWG".
motor_wire_index	Position ("1" based) in (cadr _#data) for wire type data. If nil or blank then it defa first conductor entry in the list if no override found on marker block or carried in th block's spreadsheet entry.
options	1's bit set = if new layer created, do not mark it as a "wire" layer. 2's bit set = filter out entries in (cadr _#data) for wire type data. Only process tho with the 2's bit set in the last element of each wiretype sublist. Example: (cadr _#data) = (("1AWG" "BRN_1AWG" 1) ("1AWG" "BLK_1AWG" 1) ("1AWG" "GRY_1AWG" 1) ("6AWG" "GRN_6AWG" 3))) ... means filter out all but the last entry. This is used to find the "ground" wiretype list of passed wire types. 2's bit set = don't filter out entries in (cadr _#data). Process the list "as passed." 4's bit set = process only marked wire segment. 4's bit clear = process marked "network."
misc1_spreadsheet	Any string value embedded in this API call's third argument as shown in the spread

Remarks

How this works

A marker block is placed on top of a wire segment in the drawing template and its CODE attribute value points to an entry in the ace_circuit_builder.xls spreadsheet. This entry inclu (c:ace_cb_set_wiretype...) API call.

If the marker block carries no MISC1 attribute value and the "misc1_spreadsheet" argument is blank in; the spreadsheet, then the "default_laynam" value pulled from the first subli

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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_cb_stretch_wire_connect
[Section L - Circuit Builder](#) [Send Feedback](#)

Description

Find wire under "xy" point and try to stretch to connect to nearest crossing wire or component wire connection point.

AutoLISP

(c:ace_cb_stretch_wire_connect xy options misc1_spreadsheet)

Parameters

Parameters	Description
xy	nil = use coordinate of current marker block being processed. coordinate = stretch underlying wire found at this XY coordinate. non-blank string = trim user under marker block(s) that are marked with this strin CODE attribute value (wild card match) AND have same ORDER value as the main block being processed.
options	1's bit set=suppress stretched wire from matching layer of any wire it connects to
misc1_spreadsheet	any string value embedded in this API call's second argument as shown in the spre This value is overridden by any MISC1 attribute value carried on the marker block. attribute value pulled from a global variable (if exists).

Module

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AutoCAD Electrical 2025: AutoLISP Reference

c:ace_cb_trim
[Section L - Circuit Builder](#) [Send Feedback](#)

Description

Trim wire found at location "xy".

AutoLISP

(c:ace_cb_trim xy options)

Parameters

Parameters	Description
xy	nil = use coordinate of current marker block being processed. Coordinate = trim underlying wire at this. Non-blank string = trim user under marker block(s) that are marked with this strin CODE attribute value (wild card match) AND have same ORDER value as the main block being processed.
options	1's bit set = don't trim away any IEC-tee symbols at end of trimmed wire 2's bit set = don't trim past any gap/loop, leave loop in place if exists

Module

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Description

Generate and insert a wire number on target wire network line segment "wen" or on line segment under coordinate "passed_xy".

AutoLISP

```
(c:ace_cb_wnum passed_xy wen options anno_str)
```

Parameters

Parameters	Description
passed_xy	XY position on wire segment; nil = use "wen" argument
wen	entity name of wire segment of network; nil = use "passed_xy" coordinate to find segment
options	1's bit set = insert FIXED wire number 2's bit set = return "handle" of inserted wire number (otherwise return ent name) 4's bit set = attempt to position new wire number at the passed xy coordinate bit clear = new wire number inserts at network default position
anno_str	nil for none; otherwise list of attrib = value pairs and/or other override flags

Returns

nil if fail, otherwise returns entity name (or handle) of inserted wire number

Module

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